

Events at IIIT-D

The ECE Department is organizing a talk by Prof. Aryan Kaushik, Manchester Met, UK.

The ECE Department is organizing a talk by Dr. Sourabh Upadhyay, R&D Engineer, Novanta.

The ECE Department is organizing a talk by Prof. Uday Khankhoje, Associate Professor, IIT Madras.

IIIT Delhi Signal Processing Student Chapter and IEEE Signal Processing Society Synthetic Aperture Technical Working Group are proud to host the 2025 Summer School on Synthetic Aperture Imaging for Radar, Sonar, and Optics from April 1st to 3rd, 2025, in the IIIT Delhi campus as a part of the Delhi Satellite Event for ICASSP 2025. The summer school is open to all! Registration will be opened on 26th February 2025.

The ECE Department is organizing a talk by Prof. Eiji Oki, Professor, Kyoto University

The ECE Department is organizing a talk by Dr. Naveen Mysore Balasubramanya, Associate Professor, IIT Dharwad

The ECE Department is organizing a talk by Prof. Zilong LIU from School of CSEE, University of Essex, on 10 Jan 2025 from 4 PM.

The ECE Department is organizing a talk by Devansh Gupta, PhD student at the

University of Southern California, on 8 Jan 2025 from 3 PM

The ECE Department & Wirocomm lab is organizing a talk by Prof. Pei Xiao, University of Surrey on Title: Integrated Terrestrial-Satellite Networks: Fundamentals and Future Directions

The ECE Department are organizing a talk by Prof. Maxime Guillaud, Senior researcher, Inria, MARACAS team Lyon, France on Title: Towards Practical Massive Random Access

IEEE SPS society and ECE Department are organizing a talk by Dr. Ananya Sen Gupta on Interpreting geometric concepts into sonar feature representations: case studies and challenges

"IEEE CASS Delhi Chapter and ECE Department are organizing a talk by Mr. Ang Boon Chong, Technical Lead, Intel, Malaysia and IEEE senior member".

The ECE Department is organizing a talk by Dr. Sanat K Biswas, IIIT-Delhi, on Tracking untracked space debris after an inelastic collision using physics informed neural network".

Welcome to Department of Electronics & Communications Engineering

The ECE Department's mission is to carry out advanced research and development in various areas of Electronics & Communication Engineering with different application domains, and to train and educate, at both undergraduate and postgraduate levels, engineers of outstanding ability who can become innovators and new product creators.

The vision of the department is to be a world class department which:

News and Events

IIIT Delhi Signal Processing Student Chapter and IEEE Signal Processing Society Synthetic Aperture Technical Working Group are proud to host the 2025 Summer School on Synthetic Aperture Imaging for Radar, Sonar, and Optics from April 1st to 3rd, 2025, in the IIIT Delhi campus as a part of the Delhi Satellite Event for ICASSP 2025. The summer school is open to all! Registration will be open on 26th February 2025.

Paper titled "Demonstration of Broadband Data Link over off-the-shelf Solar Module with White LED " by Rahul ., Prof. Anand Srivastava, Prof. Vivek Ashok Bohara, Dr. Deepak Solanki, and Dr. Abhijit Mitra has been accepted in IEEE/OPTICA PUBLISHING GROUP Journal of Lightwave Technology.

IIIT Delhi's ECE Department hosted a Winter School Program from December 31, 2024, to January 3, 2025, which was an engaging and inspiring experience for 9th and 11th-class students from Sardar Patel Vidyalaya and Birla Vidya Niketan schools based in Delhi.

We are pleased to announce that iHub Anubhuti's supported PI, Ranjitha Prasad, and her student's paper titled "Continual Federated Learning With Aggregated Gradients" (C-FLAG)" has been awarded the Best Poster Paper Award at the 26th International Conference on Distributed Computing and Networking (ICDCN), held at IIT Hyderabad, India, 2025!

Best Demo Award at COMSNETs 2025! Excellent work and demonstration by Jeet Tekchandani (Dual Degree Student, IIIT Delhi) and Jai Mangal (PhD Student, IIIT Delhi)

at Algorithms to Architecture, IIIT Delhi in collaboration with Shobha Ram.

Paper Title: Enhancing Wireless PHY with Adaptive OFDM and Multi-Armed Bandit Learning on Zynq System on Chip

Paper titled "L10-FeAu Perpendicular Magnetic Tunnel Junction: A DFT+NEGF Study" by Sheikh Rouf and Dr. Ram Krishna Ghosh from Quantum Electronics Lab has been published in IEEE Transactions on Magnetics.

IIIT Delhi showcased a hardware demonstration of innovative projects, "SolarComm: High-Speed optical data transmission over the solar panel" and "Blending Light and Radio: A hybrid LiFi-WiFi network for future Indoor connectivity" skilfully demonstrated by Rahul . And Saswati Paramita from WiROCOMM research group and CoELiFi Lab, led by Prof. Vivek Ashok Bohara Ashok Bohra and Dr. Abhijit Mitra.

Paper titled: Smart Element Allocation Strategies for Dynamic Optical IRS in Underwater Wireless Communication by Rehana Salam, Vivek Ashok Bohara, and Anand Srivastava is accepted for publication in the IEEE Transactions on Vehicular Technology-Oct. 2024.

A paper titled "Experimental Validation of Solar Panel Communication with Various Phase Shift Keying Modulation Schemes for Light Communication," by Rahul ., Khagendra Joshi, Prof. Vivek Ashok Bohara, Dr. Abhijit Mitra, Prof. Anand Srivastava and Deepak Solanki has been accepted for presentation in IEEE ANTS, December 15-18, 2024, in Assam, India.

Paper titled " Visible light active $0.95\text{KNbO}_3\text{-}0.05\text{Ba}(\text{Nb}_{1/2}\text{Sc}_{1/2})\text{O}_3$ ferroelectric for

enhanced photocatalytic activity" by Devender Takhar (JNU), Ankit Chahar (JNU), Deepanshu Sharma (IITD), Ram Krishna Ghosh (IIITD), and Balaji Birajdar (JNU) is published in Wiley Advanced Sustainable Systems

The Paper titled "Exploiting Side Information for Improved Online Learning Algorithms in Wireless Networks" by Dr Manjesh Kumar Hanawal (Faculty, IIT Bombay) and Dr Sumit J Darak (Faculty, IIIT Delhi) has been accepted in IEEE Transactions on Wireless Communications (TWC).

Paper titled "Impact of Non-Idealities on the Behavior of Probabilistic Computing: Theoretical Investigation and Analysis" by Amina Haroon, Dr. Ram Krishna Ghosh, Dr. Sneha Saurabh is published in IEEE Transactions on Circuits and Systems.

Paper titled "Flow-based Rate Maximization for Link Aggregation Enabled Hybrid LiFi-WiFi Network" by Saswati Paramita, Dr. Arani Bhattacharya, Prof. Vivek Ashok Bohara, Dr. Rizwana Ahmad, Ph.D., Prof. Anand Srivastava is accepted in IEEE Transactions on Vehicular Technology.

A paper titled "Digital-Twin Enabled Time Ahead Resource Allocation for Integrated Fiber-Wireless Connected Vehicular Network," by Akshita Gupta, Saurabh Jaiswal, Martin Maier, Vivek Ashok Bohara, and Anand Srivastava, has been accepted to IEEE Journal of Communications and Information Networks, Sept 2024.

Paper titled "Performance Analysis of THz and VLC Systems Under the Impact of Weather and Human Blockages," by Hanshita Prabhakar, Saumya Chaturvedi Chaturvedi, Akshita Gupta, and Vivek Ashok Bohara has been accepted for presentation

in IEEE TENCON, December 1-4, 2024, in Marina Bay Sands, Singapore.

A paper titled "Adaptive Traffic Signal Cycle Control for Green City Traffic Management," by Akshita Gupta, Anant Agarwal, and Vivek Ashok Bohara, has been accepted for presentation in IEEE TENCON, December 1-4, 2024, in Marina Bay Sands, Singapore.

Paper titled "Driving Mode Advisory for Emergency Manueuering in TransVerse Enabled Connected Vehicular Network," by Akshita Gupta, Anand Srivastava, Vivek Ashok Bohara, and Martin Maier has been accepted to IEEE Transactions on Systems, Man and Cybernetics: Systems, September, 2024.

Paper titled "Reconfigurable Radar Signal Processing Accelerator for Integrated Sensing and Communication System" by Aakanksha Tewari (PhD Student), Shragvi Sidharth Jha (BTech'23 alumni), Akanksha Sneha (PhD Student), Dr. Sumit J Dhar, Dr. Shobha Ram is accepted in IEEE Transactions on Aerospace and Electronic Systems (IEEE TAES).